

THE 25 ROBOT COMMANDMENTS

PR-25: Candidate Assurance Requirement Families for Robot Actuation

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NO UNADMITTED CONSEQUENTIAL ACTION

A proposed requirement for commanded actuation, not a declaration that every physical hazard has been eliminated.

Abstract

PR-25 organizes robot-actuation assurance into 25 candidate requirement families. It specifies what an application must establish about hazards, evidence, commands, physical constraints, protective responses, security, and lifecycle changes before granting or continuing normal actuation authority. It separates evidence validity, properties of a model, and assurance for an actual deployment.

The contribution is an implementation-neutral requirements framework, a bounded reference-model verification, and an application validation plan. Each family has an obligation, representative evidence, and a challenge that could expose a failure. Application profiles must turn these families into concrete predicates, operating limits, deadlines, and verification obligations. Their number and decomposition remain revisable.

Publication scope

This edition presents requirements, an executed finite reference-model verification, analytical examples, and open physical-assurance obligations. The reference model has 3,078 reachable states and 83,106 event transitions; the strengthened verification detects 42 selected faulty variants. These are logical reference-model results, not simulation, hardware-validation, certification, or commercial-implementation results. This edition does not establish completeness, minimality, novelty, or a particular implementation's compliance. "Shall" states a proposed conformance obligation, not a claim that an implementation already satisfies it.

Reader's route: foundations and terminology (§§1–2); admission and continued validity (§§3–4); physical assurance (§5); all 25 families (§6); conformance information (§7); worked example (§8); validation and open obligations (§9); release scope (§10).

1. Established foundations and the contribution

1.1 Prior work

Runtime assurance, shielding, formal control, and hazard analysis provide established foundations. The NASA runtime-assurance framework makes monitoring intervals and safe takeover conditions explicit. Shield synthesis addresses enforcement of selected reactive-system properties. Control barrier functions and their measurement-robust extensions provide model-dependent ways to reason about constraint preservation. None makes a Boolean admission decision a physical safety proof. [1–4]

STPA examines hazards from providing a control action, omitting one, incorrect timing or order, and inappropriate duration. PR-25 therefore addresses continuation and required protective behavior, not only permission to start. SOTER provides robotics-specific prior work on composing advanced and safe control through runtime assurance. [5–6]

PR-25's proposed contribution is the organization of these concerns into requirement families with explicit applicability, evidence, response obligations, and falsification criteria. This paper does not establish research priority or superiority over those methods.

1.2 Standards and guidance: scope, not compliance

The following are relevant reference points, not a compliance checklist. This paper uses their public official descriptions, not a clause-by-clause assessment of the normative texts. Applicability and the required assurance level depend on the actual product and use.

Reference	Relevance and limit
ISO 12100:2010 [10]	Machinery risk assessment and risk reduction. Admission logic does not replace hazard elimination or physical protective measures.
ISO 10218-1:2025 and -2:2025 [11–12]	Industrial robots and their applications/cells. The two scopes are distinct; neither covers all kinds of robots.
ISO 13849-1:2023 [13]	Safety-related control-system design. A PR-25 verdict is not a performance level or an assigned safety integrity level.
ISO 3691-4:2023 and ISO 13482:2014 [14–15]	Reference scopes for driverless industrial trucks and personal-care robots respectively. They are not interchangeable application baselines.
UL 4600 overview [16]	Evidence-based safety-case thinking for autonomous products; cited conceptually, not as a compliance or latest-edition claim.

NIST guidance covers operational-technology security and firmware resilience. RFC 9334 distinguishes attestation evidence, appraisal, and relying-party decisions. These support the treatment of provenance, evidence freshness, and recovery integrity without prescribing PR-25's implementation. [7–9]

Edition control: references identify the editions consulted. Deployment must establish its applicable standards, amendments, and assessment scope separately. Drafts are not treated as published replacements.

2. Terminology and enforcement phases

2.1 Families are not interchangeable invariants

A requirement family can contain a state predicate, a history-dependent condition, a bounded temporal response, a cross-execution information-flow obligation, or a design-time assurance argument. These are not 25 interchangeable sensor checks. A profile can instantiate more than 25 individual obligations.

Profile means an approved application description that fixes the embodiment, tools, environment, losses, hazards, assumptions, operating modes, evidence policy, relevant faults and attacks, and authority boundary.

Approval means approval by the identified responsible authority under that profile's process; it does not imply external certification.

Consequential actuation includes starting, sustaining, redirecting, or ending physical output with a credible hazard pathway. The hazard analysis also covers omission of required protective output. **Normal authority** permits a task within its admitted conditions. **Protective authority** permits only the separately justified protective behavior and its defined envelope.

2.2 Obligations apply at specific phases

Phase	Required meaning
Release	Establish the profile, hazards, evidence adequacy, control arguments, and approved configuration before the claimed use.
Admission	Evaluate the obligations due before granting authority or beginning the specified action.
Continuous	Maintain or re-evaluate obligations during execution at a justified monitoring cadence and response bound.
Completion/update	Establish physical outcomes and validate changes, restoration, or the dependencies of subsequent actions.

A family may apply in several phases. A release argument may be represented at runtime by current, dependency-bound evidence rather than re-proving it at each sample. "Continuous" does not require infinitely fast sampling; the application must justify its intervals and behavior between checks.

A future completion condition is specified before an action, but its eventual satisfaction is not assumed at admission. Reporting completion requires the specified outcome evidence. Temporal obligations require a controller or mechanism argument and deadline evidence; a current PASS alone does not prove a future response.

2.3 Applicability is controlled, not optional

The approved profile determines applicability by family, obligation, mode, command class, and phase. Every family receives a disposition and rationale. An unresolved disposition, a missing required obligation, or an unapproved exception blocks normal admission. An empty required admission set does not pass by vacuous conjunction. Selecting one convenient obligation is not a substitute for a justified applicability analysis.

NOT_APPLICABLE is a disposition, not a fifth verdict or a caller-controlled override. An inapplicability decision must itself remain valid when its dependencies change.

3. Proposed normative core

3.1 PR25-ADM-01: commanded-actuation admission

Within its declared operating domain, fault model, and trust boundary, an implementation shall initiate or continue commanded consequential actuation only under a currently valid normal or protective contract. Loss of normal admission shall prevent new normal grants and trigger the profile-defined protective response within its justified deadline. Any output during handover shall be covered by the protective transition's contract.

This is a commanded-output requirement. It is not a claim that software prevents gravity, stored-energy release, broken mechanisms, external impacts, or every uncommanded motion. The profile must allocate safeguards for those hazards.

3.2 Normal admission predicate

Let P be the profile, m the mode, a a bounded proposed command, t the evaluation time, and $R(P,m,a,t)$ the set of hard obligations whose satisfaction is due at this decision point. R includes validity of release evidence and readiness for required monitoring and protection, where the profile requires them.

```

NormalEligible(a,t) :=
  ProfileApprovedAndValid(P)
  AND ApplicabilityResolved(P,m,a,t)
  AND ContextAndEvidenceValid(a,t)
  AND EnforcementHealthy(t)
  AND ProtectiveResponseReady(a,t)
  AND R(P,m,a,t) is nonempty
  AND for every r in R(P,m,a,t): verdict(r,t) = PASS

```

`NORMAL_GRANT(a,t) implies NormalEligible(a,t)`

These names denote requirements to be implemented and justified, not self-proving Boolean inputs. The predicate establishes eligibility, not an obligation to execute. Optimization and scheduling occur only within the admitted conditions; a failed hard obligation cannot be compensated by performance elsewhere.

3.3 Evidence verdicts

Verdict	Meaning	Normal authority
PASS	Required evidence is admissible, current, sufficient and mutually consistent for the identified predicate, and that predicate is satisfied.	Eligible only if every other condition also passes.
FAIL	Valid evidence establishes a violation, or an explicit integrity or security check establishes rejection.	Denied.
UNKNOWN	Evidence is missing, insufficient, contradictory, untrusted, uninterpretable, or evaluation cannot finish within its budget.	Denied.
STALE	A previously usable observation, evidence item, or dependency has expired or been invalidated.	Denied.

Preserve individual reasons and defects. A presentation ordering is not a physical safety ranking. Evaluator failure, unsupported inputs, numerical invalidity, overflow, or timeouts must not yield PASS. A supplied verdict list does not substitute for the profile's required evidence and evaluation. Protection is admitted separately; it cannot be an unrestricted exception to this rule.

4. Decision validity and continued execution

4.1 Public requirements, not a component blueprint

Any implementation claiming this contract must identify and assess every in-scope means of commanding the relevant actuators, including autonomous tasks, teleoperation, maintenance, diagnostics, and configuration changes. There must be evidence that an unapproved path cannot bypass the claimed enforcement boundary. This is an externally assessable obligation, not a prescribed service topology, protocol, or permit format.

The authority decision must identify the subject, command meaning, relevant configuration, scope of authority, admissible state envelope, evidence dependencies, timing interval, and permitted effects. The representation is implementation-defined. The claim must remain valid at the point where output can affect the actuator, not merely at a remote planner or interface.

4.2 Evidence describes the world; it does not replace it

Origin and integrity do not prove measurement truth. An authenticated incorrect measurement remains incorrect. Acquisition time must remain distinguishable from transmission, receipt, and appraisal time. Repackaging an old observation with a new timestamp must not refresh it. The profile must justify clocks, age bounds, ordering, dependency invalidation, and uncertainty. [7]

A decision can remain valid over a bounded state envelope rather than only one exact earlier snapshot. Its current uncertainty set, mode, authority, relevant configuration, and dependencies must still satisfy the contract at execution and throughout the admitted horizon.

4.3 Three deadlines, three obligations

Start deadline: latest allowed initiation. **Control horizon:** interval for which the contract covers continued output. **Completion deadline:** latest required achievement of the defined outcome, where one is specified. Beginning before the start deadline does not authorize output beyond the control horizon. A longer task requires covered continuation or an admitted protective transition.

Changes in tool, mode, authority, configuration, ownership, evidence, or relevant state must invalidate dependent normal authority within the bound included in the physical argument. Zero propagation delay, instantaneous shutdown, and infinitely fast monitoring cannot be assumed. Recovery feasibility must include output already in flight, computation and communication delay, arbitration, actuator response, and inter-sample behavior.

4.4 Effects and results

The application must specify how retries, out-of-order delivery, ambiguous acknowledgments, power loss, and restoration affect command validity and physical effects. “Accepted” does not mean “completed.” A missing acknowledgment does not justify blindly repeating an irreversible action. At-most-once authorization is not an exactly-once physical-effect guarantee.

Child-command refinement must preserve the parent's allowed behavior under an explicit abstraction between representations. In a common predicate vocabulary, adding inherited requirements cannot weaken the parent's constraints. Units, frames, targets, and limits must survive refinement; raw set inclusion across unrelated representations is not a proof.

5. Physical constraints, uncertainty, and recovery

5.1 Three assurance claims

Admission integrity: commanded output stays within a valid normal or protective contract, assuming the claimed enforcement boundary works as specified.

Modeled physical preservation: admitted execution and takeover preserve stated physical constraints under identified plant, environment, timing, numerical, and fault assumptions.

Application assurance: implementation analysis, measurement, verification, and deployment evidence justify those assumptions and the residual-risk judgment for the actual use. Success at the first claim does not establish the other two.

5.2 Recoverability must hold through takeover

Let S denote the modeled hard-constraint set and K the region from which the designated protective policy can maintain S under the declared conditions. Require K to be contained in S . Normal segments must remain within S and preserve feasible handover states in K over the complete admitted horizon, including all response delays.

A protective path may leave the nominal task envelope while remaining within S . Calling a motion “recovery” does not authorize a hard-constraint violation. Where protection can only reduce loss, report loss mitigation rather than guaranteed preservation. A minimum-risk objective needs a declared set of alternatives and comparison criterion; it does not imply a proven global optimum.

5.3 Bounded response is not equivalent to power removal

Protection may require braking, balancing, retaining or lowering a load, cooling, or isolation. HOLD needs its own control behavior, uncertainty-growth bound, duration, exit condition, and deadline. Startup, self-test failure, shutdown, and restart require profile-specific treatment; immediate de-energization is not universally safe.

A protective contract declares its own prerequisites and covered behavior; it must not depend on recursively obtaining another mission-style recovery approval. Readiness cannot be inferred merely from denial of normal authority. A protective action must remain available under the failures it is intended to handle. Mission permission expiry, unavailable routine logging, or loss of a communication path must not block protection where the profile claims tolerance of those failures. Independence is relative to identified common causes, not merely a count of controllers. These requirements align with the timing and fallback concerns of runtime assurance. [1,6]

5.4 Uncertainty is part of the claim

Measurement, estimation, prediction, and assumptions must remain distinguishable. A statistical confidence or coverage set is not a guaranteed physical-state enclosure. State its calibration population, dependence, operating domain, and per-action or mission-horizon meaning. Where valid event-failure bounds δ_k exist, the probability of any such event is at most their sum; the union bound does not require independence. The bounds and their applicability still need justification.

Control barrier functions or reachable-set analysis can support a profile's physical argument. Mathematical regularity, actuator feasibility, sensing error, sample intervals, numerical error, and disturbances must be covered by the chosen argument. Naming a technique does not discharge these obligations. [3–4]

6. Revised PR-25 registry

Foundation: declared hazards and configuration

All entries are candidate requirement families, not claims of implemented functions. The obligation, evidence, and challenge are retained from the v0.2 registry. The phase focuses and boundary notes below are v0.4 editorial clarifications: illustrative allocations, not a fixed schedule for every embodiment. Each approved profile must resolve its actual phases. No challenge in this section is reported as executed.

PR-01 — Hazard-control traceability and closure

Phase focus: Release; admission; completion/update.

Every declared relevant hazard shall have a disposition, safety requirements, responsible owner, control allocation, verification evidence, and explicit residual-risk treatment. Admission checks the current applicability and status of that package, not whether all possible hazards have been discovered. **Evidence:** versioned hazard log and requirement-control-test links. **Challenge:** introduce a payload drop hazard with no allocated control; reject profile release or affected operation. This narrows “hazard closure” to auditable declared scope.

Boundary: Traceability of the declared hazard set is not evidence that every possible hazard has been identified.

PR-02 — Assumption and operating-domain validity

Phase focus: Release; admission; continuous.

The contract shall list assumptions, distinguish monitored assumptions from design-justified assumptions, and define responses to invalidity or uncertainty. **Evidence:** domain limits, monitored conditions, dependency certificates, boundary margins. **Challenge:** introduce an unqualified floor condition or loss of required visibility. Mission continuation must lose admission before recoverability is lost; undetectable assumptions need separate assurance.

Boundary: Loss of an unmonitorable assumption cannot be treated as guaranteed detectable; its justification and residual limitation must be explicit.

PR-03 — Identity and configuration integrity

Phase focus: Release; admission; continuous; completion/update.

Physical robot, tool, controller, calibration, payload class, and active software shall match the approved configuration. **Evidence:** authenticated configuration measurements and physical tool/payload confirmation where needed. **Challenge:** swap an end effector after permit issue. The old permit must not authorize the new tool. A software identifier alone cannot establish the physical attachment.

Boundary: Configuration validity must connect recorded identity to the actual physical attachment and condition relevant to the claim.

Foundation: time, evidence, and state

PR-04 — Temporal coherence and freshness

Phase focus: Release; admission; continuous; completion/update.

Required observations, estimates, commands, and permits shall satisfy declared clock, age, skew, and ordering constraints. **Evidence:** acquisition time, clock identity, timing-error bounds, age budgets, watchdog state.

Challenge: freeze a sensor while publishing new packet timestamps, or roll the wall clock backward. Neither condition may refresh the underlying measurement.

Boundary: A current signature or fresh transport packet does not establish a current physical observation. [7]

PR-05 — Evidence provenance and integrity

Phase focus: Release; admission; continuous; completion/update.

Safety evidence shall have an admissible origin, intact content, and a traceable dependency chain. **Evidence:** authenticated source and artifact identities, derivation versions, integrity verification, revocation state.

Challenge: alter a calibration artifact or inject planner-created evidence under a sensor identity. Rejection of provenance is separate from PR-06's treatment of uncertainty.

Boundary: The profile must justify its provenance and integrity mechanisms; it cannot waive required evidence merely to obtain PASS. Integrity does not establish the truth or adequacy of the source observation.

PR-06 — Epistemic type and uncertainty integrity

Phase focus: Release; admission; continuous; completion/update.

Measured, estimated, predicted, assumed, and externally asserted values shall remain distinguishable. Derived evidence shall retain relevant uncertainty and correlation information. **Evidence:** typed evidence, error bounds, validation domain, dependence declarations. **Challenge:** label a predicted clear corridor as a current observation, or count correlated estimates as independent confirmation. Admission must not gain unjustified confidence.

Boundary: Any probabilistic coverage used by an obligation must retain its assumptions and nonzero tail; it is not silently upgraded to an absolute bound.

PR-07 — State-estimation consistency

Phase focus: Release; admission; continuous.

The state estimate used for control shall satisfy declared consistency checks and provide an adequate bound for the safety calculation. **Evidence:** estimator residuals, sensor compatibility checks, plausibility limits, conservative state sets. **Challenge:** wheel slip creates conflicting motion estimates. A plausible point estimate must not pass when its uncertainty exceeds the available safety margin. Consistency alone does not prove correctness.

Boundary: Error bounds and consistency tests require validation against the intended domain, not only agreement among sensors sharing a common defect.

Command: authority, meaning, and effects

PR-08 — Authority, purpose, and delegation binding

Phase focus: Release; admission; continuous.

The command shall be authorized for its target, purpose, operating mode, and current authority epoch.

Evidence: scoped authority, delegation chain, policy decision, local revocation state. **Challenge:** present an otherwise valid permit after a local authority-epoch change. Safety and authorization remain separate: an authorized command can still be physically unsafe.

Boundary: The independently justified protective contract has its own bounded authority; it is not a general bypass of authorization.

PR-09 — Command semantic determinacy

Phase focus: Release; admission; continuous; completion/update.

The action shall have one validated interpretation at the actuator boundary, including units, reference frame, target, limits, and completion criteria. **Evidence:** canonical command encoding, schema version, unit/frame checks, reference resolution. **Challenge:** interchange millimeters and meters or change the frame after evaluation. Natural-language intent must be resolved before admission.

Boundary: Completion criteria are defined at admission and assessed when the outcome can be observed; future completion is not presumed.

PR-10 — Command freshness, replay, and effect integrity

Phase focus: Release; admission; continuous; completion/update.

The actuation interface shall reject expired, reused, reordered, or semantically altered permits according to the declared effect model. **Evidence:** command IDs, sequence scope, durable consumption records or reconciled physical state. **Challenge:** lose an acknowledgment, reboot, then retransmit a gripper-open action. The interface must not blindly repeat an uncertain physical effect.

Boundary: Authorization and outcome semantics must be separated. At-most-once acceptance does not alone establish exactly-once physical execution.

Command: transitions, ownership, and interruption

PR-11 — Preconditions and transition binding

Phase focus: Release; admission; continuous; completion/update.

A command shall start only from the required discrete preconditions and admitted continuous state envelope; its allowed transition and completion predicate shall be specified. **Evidence:** current mode, interlocks, state/version checks, trajectory envelope, postcondition definition. **Challenge:** a support fixture is removed between checking and execution. PR-11 defines what completion requires; PR-22 governs whether the outcome is truthfully reported.

Boundary: A runtime check must cover the conditions due now. It does not label an unobserved future postcondition as already satisfied.

PR-12 — Resource ownership and arbitration

Phase focus: Release; admission; continuous.

Every controlled actuator or shared safety resource shall have coherent ownership with defined precedence and atomic handover at the appropriate boundary. **Evidence:** ownership and fencing epochs, arbitration rules, reservations, local enforcement state. **Challenge:** two controllers hold inconsistent distributed leases. Only the locally valid owner may issue normal outputs; the protective controller's priority must remain effective.

Boundary: A locally enforceable arbitration decision and its response time need evidence; a conceptual ownership rule alone is insufficient.

PR-13 — Bounded interruption, preemption, and required response

Phase focus: Release; admission; continuous; completion/update.

Cancellation and protective requests shall receive the defined response within a justified bound without unsafe interruption or hazardous omission. **Evidence:** temporal automata, deadlines, handover protocol, progress watchdogs. **Challenge:** cancel while carrying a load or suppress a required brake request. Verify the bounded protective behavior rather than merely an API acknowledgment.

Boundary: An acknowledged cancellation is not evidence of timely protective actuation. The measured response must match the declared temporal obligation.

Physical safety: capability, envelopes, and exposure

PR-14 — Capability and reserve sufficiency

Phase focus: Release; admission; continuous.

The planned motion and its protective response shall fit validated actuator, energy, thermal, computation, and communication capabilities. **Evidence:** derating model, load/payload estimate, reserve budgets, deadline margins. **Challenge:** reduce available braking torque or recovery energy after planning. Nominal task feasibility without recovery reserve must not pass.

Boundary: Useful normal operation and the complete protective transition must both fit the qualified operating and reserve bounds.

PR-15 — Safe-envelope and recoverability preservation

Phase focus: Release; admission; continuous.

The admitted control shall preserve hard physical constraints and retain fallback feasibility over its complete bounded execution and takeover horizon under the declared model. **Evidence:** verified envelope, reachable-set bound, robust barrier or equivalent control argument. **Challenge:** choose a trajectory that is presently collision-free but crosses the last safe braking boundary before the next evaluation.

Boundary: The claim covers all behaviors allowed by the contract under its assumptions, not only a nominal planned trajectory.

PR-16 — Protected-entity and exposure constraints

Phase focus: Release; admission; continuous.

The profile shall identify protected humans and other entities and enforce the applicable contact, separation, force, exposure, and access constraints. **Evidence:** protected-space model, sensing capability, application-specific limits, contact classification. **Challenge:** a hand enters a pinch region while all joint limits remain satisfied. Intended physical contact needs qualified limits, not a universal no-contact rule.

Boundary: The application must justify who or what is protected and how; contact may be intended but still requires qualified constraints.

Physical safety: execution, protection, and recovery

PR-17 — Command-actuation coherence

Phase focus: Release; continuous; completion/update.

Actual actuator behavior shall remain within the admitted command and tracking-error bounds; material divergence shall produce the specified response. **Evidence:** independent or appropriately justified feedback, tracking monitors, actuator diagnostics. **Challenge:** a drive continues accelerating after a braking request. A successful command acknowledgment is not evidence of successful actuation.

Boundary: Uncommanded motion may require safeguards beyond the command channel. The application must identify that limitation rather than infer control from authorization.

PR-18 — Runtime monitor and protective authority integrity

Phase focus: Release; admission; continuous.

The monitor, arbitration, and protective controller shall retain effective authority and meet their integrity and timing requirements under the declared faults. **Evidence:** architectural independence argument, watchdogs, execution-time analysis, bypass/interference tests. **Challenge:** saturate the autonomy processor or compromise its process. Demonstrate that the protected path still enforces its allocated safety function.

Boundary: The relevant fault model must include common dependencies and bypasses. A failed enforcement mechanism cannot prove its own integrity by reporting PASS.

PR-19 — Assured recovery and minimum-risk transition

Phase focus: Release; admission; continuous; completion/update.

An identified protective policy shall be feasible from all admitted handover states and reach its specified condition within the required bounds, where reachability is claimed. **Evidence:** recovery envelope, resource reserves, transition controller and terminal-state criteria. **Challenge:** lose mission communication during motion. “Minimum risk” must specify the candidate responses and objective; do not imply a proven global optimum without one.

Boundary: When recovery premises are lost, report the loss of the assurance claim and apply the separately justified mitigation strategy; do not invent a safe fallback.

Resilience: faults, trust, and truthful outcomes

PR-20 — Fault containment and common-cause resilience

Phase focus: Release; admission; continuous.

The design shall meet its allocated safety functions under the declared fault combinations and constrain propagation between components. **Evidence:** fault analysis, dependent-failure analysis, isolation architecture, fault-injection results. **Challenge:** fail a shared clock, power rail, or sensor dependency. Two nominally independent monitors must not be counted as independent evidence when that fault defeats both.

Boundary: Evidence for redundancy must address shared power, clocks, sensors, communication, software, and environmental causes relevant to the application.

PR-21 — Cyber and trust-boundary integrity

Phase focus: Release; admission; continuous; completion/update.

Only authorized, integrity-protected transitions may change physical authority, evidence policy, configuration, or protected control behavior. **Evidence:** threat model, authenticated interfaces, secure boot/update arrangements, partitioning and denial-of-service analysis. **Challenge:** spoof a sensor origin or exhaust the control network. Cyber rejection must preserve the locally assured protective response.[8]

Boundary: Cybersecurity and physical protection have coupled timing and availability obligations; denying a network request is not a complete safety response. [8]

PR-22 — Outcome and audit truthfulness

Phase focus: Release; continuous; completion/update.

Reports shall distinguish accepted, started, executing, completed-and-verified, failed, interrupted, and outcome-unknown. Downstream actions shall use verified preconditions rather than optimistic success messages.

Evidence: physical feedback, outcome predicates, sequence-linked event records and explicit gaps. **Challenge:** lose feedback after motion. The outcome stays unknown; a full log device must not block urgent protection.

Boundary: The evidence needed to begin the next action must be available even when a prior action was accepted successfully but its outcome is uncertain.

Lifecycle: change, memory, and restoration

PR-23 — Controlled adaptation and update integrity

Phase focus: Release; admission; continuous; completion/update.

Changes to models, policies, maps, calibration, controllers, or evidence evaluation shall remain within an approved adaptation envelope or undergo reassessment before use. **Evidence:** signed/authorized change record, dependency impact analysis, validation results, active-version confirmation. **Challenge:** change a learned

controller online without updating its safety envelope. Reject the affected transition and invalidate dependent permits.

Boundary: A valid change signature alone does not establish behavioral compatibility or preserve the old assurance argument.

PR-24 — Memory, retention, and privacy integrity

Phase focus: Release; admission; continuous; completion/update.

Safety-relevant memory shall be typed, current, and protected; retention and access rules shall preserve required safety evidence while implementing the deployment's privacy requirements. **Evidence:** data inventory, provenance, retention/access policy, buffer capacity and erasure controls. **Challenge:** stale task memory changes a target, or a deletion operation removes the only required current safety state. Split safety-state and personal-data obligations rather than treating them as one universal retention rule.

Boundary: Privacy and safety-state duties must be separately instantiated. A conflict requires a resolved application policy, not a blanket retention rule.

PR-25 — Restore, rollback, and version coherence

Phase focus: Release; admission; completion/update.

Restore shall produce an approved compatible combination of software, models, calibration, evidence policy, and persistent state while preserving anti-replay and anti-rollback requirements. **Evidence:** compatibility manifest, security version floor, restore validation, boot epoch and consumption-state reconciliation. **Challenge:** restore an old signed image and stale command journal. A signed image may still be revoked or incompatible; physical effects are not undone by software rollback.[9]

Boundary: Restoring software does not undo a physical effect. The application must reconcile present physical state and persistent records before normal resumption. [9]

7. Application conformance information

7.1 Each obligation must be traceable

An application claiming PR-25 conformance must identify the public registry revision and the concrete obligations it instantiates. For each obligation, record its identifier and revision, predicate or temporal requirement, applicability rationale, enforcement phases, linked hazards, assumptions, evidence dependencies, uncertainty treatment, units, freshness and invalidation conditions, deadlines, required response, responsible owner, and verification artifacts.

This is an information requirement, not a wire schema, application programming interface, executable configuration, or implementation disclosure. The representation and control architecture are implementation-defined. A conformance argument must remain reviewable in the declared assessment setting.

7.2 Conformance record: illustrative contents

Record element	What the assessor needs
Scope	Embodiment, operating domain, modes, tasks, protected entities, losses, excluded conditions, and claim interval.
Applicability	A justified disposition for every family; all instantiated obligations and their due phases; explicit handling of unresolved decisions.
Evidence	Identified observations, analyses, tests and assumptions; their adequacy, integrity, dependencies and validity conditions.
Timing and response	Monitored event, justified detection and response bounds, behavior during handover, protective envelope, and restart conditions.

Record element	What the assessor needs
Results	What was checked, against which claim, using which artifact version, with which result and limitations.
Governance	Responsible owners, review authority, change control, exceptions, residual-risk decisions, and invalidation responses.

7.3 Three reports must remain distinguishable

A **requirements review** assesses whether a profile has precise, allocated and assessable obligations. A **model verification report** identifies a model, assumptions, properties, state space or proof obligations, tool configuration and results. An **application assurance report** connects implementation, plant, measurements, faults and operational evidence to the deployment claim. None substitutes for the other two.

Validation must reject absent references, mismatched evidence meanings, unresolved applicability, invalid numbers, and a representation that disguises a required failure as inapplicability. Deterministic evaluation means identical approved inputs and evaluator semantics produce the same result; it does not eliminate numerical approximation, physical variability, or mistaken evidence.

7.4 Family overlap is allowed; hidden obligations are not

PR-04 concerns time validity; PR-10 concerns replay and effects. PR-05 concerns evidence origin and integrity; PR-06 concerns its meaning and uncertainty. PR-11 specifies transitions and completion; PR-22 reports observed outcomes. PR-15 concerns admitted trajectories; PR-19 concerns protective behavior. PR-18 concerns protective authority; PR-20 concerns fault containment and common causes. These distinctions improve allocation and testing; they do not prove logical independence.

8. Worked mobile-manipulator example

8.1 Scope and assumptions

Consider a mobile manipulator approaching a stationary obstacle on a qualified level indoor floor, with its arm stowed and load mechanically retained. The following values are invented engineering assumptions, not measured performance, standard-mandated limits, or a validated robot profile. The example excludes approaching people, slopes, stairs, changing payloads, and compliant contact.

Quantity	Symbol	Illustrative value
Upper-bound speed at observation	v	1.00 m/s
Maximum forward acceleration before braking	a	0.50 m/s ²
Full observation-to-braking delay	τ	0.12 s
Minimum effective braking deceleration	b	1.50 m/s ²
Combined distance uncertainty allowance	e	0.08 m
Required residual separation	c	0.20 m
Measured distance from relevant robot boundary	d	0.90 m

Assume nonnegative forward speed until stopping, acceleration bounded by a throughout τ , and subsequent effective deceleration of at least b . Brake buildup is included in τ . A conservative required distance under these assumptions is:

```

d_required = v*tau + 0.5*a*tau^2
            + (v + a*tau)^2/(2*b) + e + c
            = 0.778133... m
margin      = d - d_required = 0.121866... m

```

This one predicate passes under the stated assumptions. It does not grant authority by itself: the applicable evidence, configuration, authority, state, and protective-readiness obligations must also pass.

8.2 A delay mutation changes the result

At $\tau = 0.30$ s, with the remaining illustrative values unchanged:

```

d_required = 0.3000 + 0.0225 + 0.440833... + 0.0800 + 0.2000
            = 1.043333... m > 0.90 m

```

The same recorded clearance is insufficient. A real system must trigger protection before its last feasible intervention point; denying normal motion at a later unsafe snapshot cannot retroactively create stopping distance.

8.3 What the example does not establish

The distance and time allowances must cover actual error sources without unjustified double counting. Moving obstacles require a closure-motion bound. A raised load changes the braking and retention assumptions. After uncertain gripper execution or restoration, current physical state must be reconciled before retrying. These are profile obligations, not conclusions of the one-dimensional arithmetic.

9. Verification and falsification plan

9.1 Match the method to the claim

Claim	Required evidence	Representative falsifier
Conjunctive admission	Evaluator tests and an identified formal admission definition.	A normal grant with a required obligation missing, unresolved or non-PASS.
No commanded-output bypass	Complete in-scope interface analysis and adversarial testing.	An in-scope source commands output without a valid contract.
Continued context validity	Races, expiry, tool changes and dependency-invalidation tests.	Normal continuation after an invalidating change outside the justified response bound.
Physical preservation	Qualified dynamics, uncertainty and control analysis, plus physical evidence.	An execution within the declared model violates a hard constraint.
Timely protection	Worst-case timing argument and measured handover under relevant faults and load.	Required protection begins or completes after its justified deadline.
Effect and outcome integrity	Retry, crash, restoration, feedback-loss and reconciliation tests.	Unauthorized repeated physical effect or completion claimed without its required evidence.
Useful operation	Representative nominal tasks and denial/intervention analysis.	Apparent success obtained only by preventing every useful action.

Use unit tests for pure predicates; formal models for admission, authority, and temporal behavior; physical-system analysis for dynamics; simulation for scenario exploration; hardware-in-the-loop work for timing and interfaces; and controlled physical testing for application assumptions. Results from one method do not automatically cover the others.

9.2 Minimum adversarial campaign

Evidence and evaluation: missing, stale, forged or contradictory evidence; stale acquisition with fresh transport timestamps; unsupported schema; NaN or overflow; unresolved applicability; empty required set; unauthorized exception; evaluator failure; budget overrun.

Command and transition: post-check tool swap; authority revocation; split ownership; changed units or frame; duplicate effect after acknowledgment loss; restoration of consumed command state; unsafe cancellation; suppressed required protective action.

Plant, protection and lifecycle: actuator mismatch; reduced braking; low recovery reserves; common sensor, clock or power failure; scheduler starvation; communication loss; raised-load interruption; unavailable logs; false completion; unapproved adaptation; memory deletion; revoked or incompatible restore.

For each case record the initial conditions, injected event, affected obligations, expected verdict, allowed outputs during response, deadlines, observed physical trace, and outcome. Include coupled faults with a shared causal pathway. The examples are proposed challenges, not an executed coverage claim.

Mutation campaigns should test whether weakened checks are detected. Report the mutant-selection process, equivalent or invalid variants, exclusions, concrete counterexamples, and remaining blind spots. Detection of selected faults is not a universal failure-coverage percentage.

9.3 Metrics and statistical limits

Report unauthorized normal releases, missing-evidence admissions, stale uses, physical-constraint violations, deadline misses, repeated effects, false success reports, and missed invalidations. Also report useful completion, false denial, unnecessary protective interventions, and time in degraded operation.

Zero observed violations is not zero risk. Under independent Bernoulli trials with a common unknown failure probability p and a pre-specified fixed number n of trials, observing zero failures gives the exact one-sided 95% upper confidence bound:

$p_{upper} = 1 - 0.05^{(1/n)}$, approximately $3/n$ for large n .

The expression follows from $(1-p)^n = 0.05$. The population, scenario distribution, exposure unit, independence, and common-probability assumptions must be justified. Correlated tests, adaptively selected scenarios, and enumeration of formal states do not automatically constitute such trials. A green test suite alone is neither this statistical bound nor a physical safety proof.

9.4 Testing coverage and the decomposition

Freeze the operating domain, failure ontology, modes, and property language before evaluating coverage. Derive unsafe-control scenarios independently of the 25 headings, then map them to concrete obligations and controls. Include omission, timing, duration, startup, maintenance, shutdown, and foreseeable misuse. A taxonomy that merely restates the headings is not independent coverage evidence. [5]

For logical independence of precisely defined predicates U_i under assumptions A , seek a modeled trace satisfying:

$A \text{ AND every other } U_j \text{ AND NOT } U_i$

First establish that the relevant assumptions and comparison model admit meaningful executions; an inconsistent model makes an implication vacuous. Unsatisfiability of the counterexample condition establishes implication only in that specified model. A timeout or unknown solver result establishes neither implication nor independence. Logical independence, separate enforcement mechanisms, and incremental reduction of loss are different questions.

Useful responsibilities need not form a minimal mathematical basis. There is no demonstrated result here that exactly 25 is necessary or sufficient for every embodiment. Counterexamples may require an entry to expand, split, merge or retire, or may require application-specific obligations outside the current list.

9.5 Executed finite reference-model verification

The accompanying reference-model package was executed with CPython 3.13.5. The supplied original checker reproduced its saved JSON byte for byte. A second, strengthened verifier preserves that transition model while

stating its authority conditions separately from the implementation's normal-admission helper. The source and results are included in `reference_model/`; the separate Verification Note defines the artifact correspondence, commands, and findings.

Check	Executed result	Interpretation
Reachable baseline graph	3,078 states; 83,106 transitions; 27 events per state	No counterexample to the strengthened logical properties in this finite graph.
Constructive state census	A concrete initial-state witness for each of the 3,078 states, plus closure on all 83,106 transitions	A separately expressed closed state set is exactly reachable, not merely a guessed count.
Transition correspondence	8,192 encodings x 27 events = 221,184 comparisons; zero mismatches	The supplied transition function agrees with a separately written arithmetic specification, including unreachable encodings.
Admission fold	6 reachable accumulator states; 72 transitions; all 271,453 sequences of lengths 0 through 5 over 12 input letters	Bounded executable equivalence checks supplement the inductive fold argument.
Selected faulty variants	27 verdict variants, 6 original supervisor variants, 9 shared-helper variants; 42 detected out of 42 selected	Sensitivity to these selected defects, not an estimated fault-coverage or safety percentage.
Invalid-code boundary	19 malformed-input cases rejected by a separate strict admission helper	Extra-domain hardening; the original model explicitly assumed valid finite codes.

The fold argument maintains three propositions for every processed prefix: all applicability dispositions are resolved; at least one obligation is required; every required verdict is PASS. The base state is resolved, empty, and all-pass. Each of the twelve valid input letters preserves these meanings, so the final conjunction equals the stated set definition for every finite sequence. This is a written induction supported by executable checks, not a proof-assistant certificate.

The original additional case total of 302 comprises 300 positional 25-entry evaluations, one all-N/A 25-entry evaluation, and one empty-sequence evaluation. It is not 302 distinct 25-entry vectors; those executions contain 278 distinct sequences in total. The 25 positions model logical slots, not the physical semantics of the 25 families.

Verification defect found and addressed. The original model and property checker both called the same `normal_ok` helper. When that helper was changed to omit the nonempty-required-set condition, the old checker still returned PASS. The strengthened oracle does not call that helper, and the regression suite detects omission of each of its nine normal conditions. This is a repair to the verification setup; it is not evidence of a fault in the unmodified baseline. Two original named supervisor variants change more than one check, so the mutation total must not be described as a count of independent single-line faults.

Model boundary. Ten trusted Boolean summaries, four permit states, and a freshness bit describe local atomic logical grants. Expiry is an invalidation event, not elapsed physical time. The model does not establish sensor truth, recovery feasibility, controller dynamics, scheduling, cryptography, persistent replay protection, distributed ownership, or physical outcomes. NORMAL and PROTECTIVE are both reachable, but task completion and availability are not measured. The checker, oracle, interpreter, and abstraction remain in the trust base; the second oracle is not independent external certification. The evidence does not establish conformance of any commercial implementation.

The worked stopping-distance example remains an analytical calculation with invented assumptions. The embodiment-level challenge cases remain unexecuted. Application assurance still requires the obligations in section 9.6.

9.6 Whole-system assurance: open proof obligations

For a specified embodiment, let S be the hard safety set and K a subset from which the designated protective policy can maintain S under the declared fault and disturbance bounds. Establish that the initial possible-state set lies in K .

For every permitted normal segment, require a conservative reachability argument covering estimation uncertainty, disturbances, sampling, numerical error, computation, communication, arbitration and actuation delay. The entire segment must remain in S , and every possible handover state must remain in K . If a contract allows several controls, the argument must cover the allowed behavior, not only a selected nominal trace.

Require the protective policy to preserve S from every admitted handover state. Include any relevant discrete transitions, mode changes, resets, and physical contact in the model. Establish that the actual implementation, sensing, scheduling, and actuator behavior refine that model within the stated error and timing bounds.

Environment assumptions must hold for the claimed interval, or their loss must be detected while an assured handover is still feasible. Assumptions that cannot be operationally monitored need a separate justification and an explicit limitation. Each assumption has an owner, evidence, scope, and invalidation response.

Conditional conclusion: the modeled physical trajectory remains in S over the covered interval. The reasoning composes admitted segments with protective transitions, using the within-segment physical argument. Unbounded-time claims additionally need time divergence or exclusion of Zeno execution. Eventual arrival in a recovery target needs a separate progress argument and justified timing assumptions; invariance alone does not establish arrival.

This is an argument outline and a list of proof obligations, not a discharged theorem for a real robot. The timing and takeover concerns are consistent with established runtime-assurance work; no certificate or proof artifact for a PR-25 application accompanies this edition. [1]

10. Public release scope and review procedure

PR-25 publishes an implementation-neutral set of candidate requirements. The framework does not depend on a particular product, provider, programming language, or implementation design. The public contract and review questions stand on their own.

The public title is **The 25 Robot Commandments**. The technical scope is **25 candidate assurance requirement families for robot actuation**. The number denotes this edition's organization, not a proof of universality or an institutional standard. No certification or endorsement by the cited organizations is implied.

A useful review submission identifies the profile and assumptions, affected family or missing responsibility, initial state, sequence of events, permitted or prohibited output, and the evidence that contradicts the claim.

Distinguish a flaw in the requirements, a violated assumption, an implementation defect, and an inadequate validation argument. Publish resolved revisions and unresolved limitations rather than forcing every challenge into the existing numbering.

v0.4 changes: retains all 25 family identifiers, titles, obligations, and challenge cases; restores finite-model evidence after executing the actual source package; adds a separately stated verification oracle and shared-helper regressions; distinguishes executed cases from distinct inputs; rejects malformed admission codes at the demonstration boundary; and preserves the separation between logical reference evidence and application assurance. The requirement-family organization is unchanged from v0.3.

References

The following bibliography is retained from the supplied public v0.3 paper. References identify particular research publications and publisher-described editions; this v0.4 artifact audit is not a new standards-status or clause-by-clause compliance assessment. They support the established foundations and scope statements, not a claim that PR-25 has been validated or endorsed. The requirement families and worked example are the proposal evaluated here.

[1] **J. Tanner Slagel, Lauren M. White, Aaron Dutle, César A. Muñoz, and Nicolas Crespo.** A Formal Verification Framework for Runtime Assurance. NASA Formal Methods, 2024. Monitoring and takeover assumptions.

[Source 1](#)

[2] **Roderick Bloem, Bettina Könighofer, Robert Könighofer, and Chao Wang.** Shield Synthesis: Runtime Enforcement for Reactive Systems. 2015. arXiv:1501.02573. Reactive-system enforcement.

[Source 2](#)

[3] **Aaron D. Ames, Samuel Coogan, Magnus Egerstedt, Gennaro Notomista, Koushil Sreenath, and Paulo Tabuada.** Control Barrier Functions: Theory and Applications. 2019. arXiv:1903.11199. Model-based control foundations.

[Source 3](#)

[4] **Ryan K. Cosner, Andrew W. Singletary, Andrew J. Taylor, Tamas G. Molnar, Katherine L. Bouman, and Aaron D. Ames.** Measurement-Robust Control Barrier Functions: Certainty in Safety with Uncertainty in State. 2021. arXiv:2104.14030. State-estimation uncertainty.

[Source 4](#)

[5] **Nancy G. Leveson and John P. Thomas.** STPA Handbook. MIT-STAMP-001, March 2018. Hazard analysis and unsafe control actions.

[Source 5](#)

[6] **Ankush Desai, Shromona Ghosh, Sanjit A. Seshia, Natarajan Shankar, and Ashish Tiwari.** Soter: Programming Safe Robotics System using Runtime Assurance. UC Berkeley technical report EECS-2018-127, 2018. Earlier technical report of the SOTER work.

[Source 6](#)

[7] **IETF.** RFC 9334: Remote Attestation procedureS (RATS) Architecture. January 2023. Evidence, appraisal, and freshness.

[Source 7](#)

[8] **NIST.** SP 800-82 Rev. 3: Guide to Operational Technology (OT) Security. September 2023. Physical and operational constraints on cybersecurity.

[Source 8](#)

[9] **Andrew Regenscheid; NIST.** SP 800-193: Platform Firmware Resiliency Guidelines. May 2018. Protection, detection and recovery.

[Source 9](#)

[10] **ISO.** ISO 12100:2010: Safety of machinery — General principles for design — Risk assessment and risk reduction. Official scope and publication listing.

[Source 10](#)

[11] **ISO.** ISO 10218-1:2025: Robotics — Safety requirements — Part 1: Industrial robots. Official scope and publication listing.

[Source 11](#)

[12] **ISO.** ISO 10218-2:2025: Robotics — Safety requirements — Part 2: Industrial robot applications and robot cells. Official scope and publication listing.

[Source 12](#)

[13] **ISO.** ISO 13849-1:2023: Safety of machinery — Safety-related parts of control systems — Part 1: General principles for design. Official scope and publication listing.

[Source 13](#)

[14] **ISO.** ISO 3691-4:2023: Industrial trucks — Safety requirements and verification — Part 4: Driverless industrial trucks and their systems. Official scope and publication listing.

[Source 14](#)

[15] **ISO.** ISO 13482:2014: Robots and robotic devices — Safety requirements for personal care robots. Official scope and publication listing. No claim that a draft service-robot revision is a published replacement.

[Source 15](#)

[16] **UL Solutions.** UL 4600 Edition 3 Updates Incorporate Autonomous Trucking. 2023. Public overview used for safety-case context, not a latest-edition or compliance claim.

[Source 16](#)

Document control: PR-25 public technical paper, v0.4, 13 September 2026. This is a candidate requirements publication, not a standards-compliance report. The executable finite reference checker and its verification results are separate files in the accompanying public package. No robot controller, application certificate, or hardware-validation attachment is implied.